1) Acoustic acquisition and hardware preparation

Week 5 closeout and Week 6 preparation
Owner: Kaan | Module A - Acquisition Hardware & Characterisation
Reporting period: 19-25 October 2026
Prepared: 26 October 2026 | Revision 1.1 | English edition

An inventory record, dimensioned mounting and phantom proposals, an ECM comparison plan and a Week 6 checklist were prepared against tasks A1-A5 [1]. These outputs document design and preparation; they are not physical test results.

Category	Current status
Completed preparation	Mechanical concepts, inspection method, repeatability procedure and conditional test plans were documented. Physical assembly was not confirmed.
Reported receipt	All items listed in the receipt table were reported received except Kingstate KECG2740PBJ, which is awaiting delivery. Quantities and acceptance inspection evidence were not supplied.
Actual experiments	NOT YET MEASURED. Kaan confirmed that no measurement had been performed because all materials were not yet available.
Open work	Kingstate delivery, item-level acceptance inspection, mechanical-part availability and fit, electrical checks and all performance measurements.

Updated receipt status: the Module A owner confirmed that all items listed on page 2 were received except Kingstate KECG2740PBJ. This supersedes the earlier partial-delivery status. Receipt does not establish inspection or measured performance. The document is dated 26 October 2026; no earlier submission is claimed.

Progress since Week 4: requirements were translated into a removable mounting concept, a dimensioned fixture, inspection fields, a ten-remount method and an executable test sequence. Requirements acceptance and measured performance remain open.

2) A1 - Receipt and inspection

The order baseline is the reported 16 October order and Week 4 handoff [4]. Receipt is distinguished from count verification and acceptance inspection. Unknown counts are not zero. Unconfirmed items are not labelled received, missing or damaged.

Item / ordered quantity	Receipt status
ESP32-S3 board / 2 [4]	RECEIVED
INMP441 module / 3	RECEIVED
Resistors / not itemised	RECEIVED
Capacitors / not itemised	RECEIVED
Jumper wires / not itemised	RECEIVED
Kingstate KECG2740PBJ / 2	AWAITING DELIVERY
MCP6022 / 5	RECEIVED
Exciter 30 mm / 12 Ω / 2	RECEIVED
PAM8403 module / 4	RECEIVED
Breadboard / other prototyping items	RECEIVED

Planned acceptance inspection

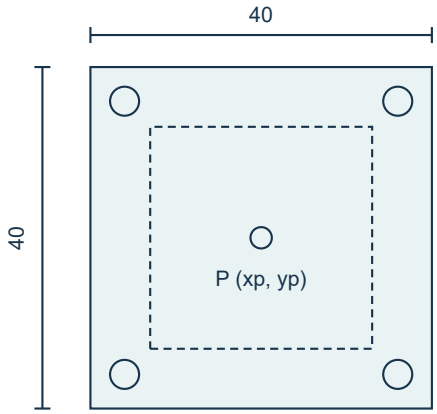
Each item will be counted against the order. Model/board markings, microphone port, headers, loose joints and visible damage will be inspected without power. Resistor/capacitor values and polarity will be recorded. Suspect parts will be isolated. Exact ESP32 and microphone board variants will be checked before pin assignment.

Evidence: item ID, ordered/received/accepted quantities, marking, condition, date, inspector and product-only photo filename. Addresses and receipts will not be published. No damage was reported; an unperformed inspection cannot establish that parts are undamaged.

3) A2 - INMP441 acoustic mounting

A removable insulated cassette linked to the existing chestpiece outlet through a short tube was proposed. The cassette will be restrained on the frame; the loading screw will act on the chestpiece rim, never on the microphone. Cable strain relief and a slack loop will limit pulling and vibration. Acoustic improvement has not been validated.

Cassette plan - proposed dimensions (mm)



4 x Ø3.4; centres 4 / 36 from edges

Dashed PCB envelope: 26 x 26 maximum

P: port position must be measured

Outer height: 12; two removable halves

Tube length: nominal 20; bore to fit

Annular gasket: nominal 1

Acoustic path / section - schematic, not to scale



Do not cover the port; support the PCB edges, not the microphone can.

Before fabrication: measure PCB outline, component/header heights, port coordinates, head size and outlet diameter. The 40 × 40 × 12 mm cassette is a conditional envelope, not the breakout datasheet size. Enlarge it if necessary. Tube bore must fit the actual outlet; the nominal 20 mm tube must not kink. [5, 6]

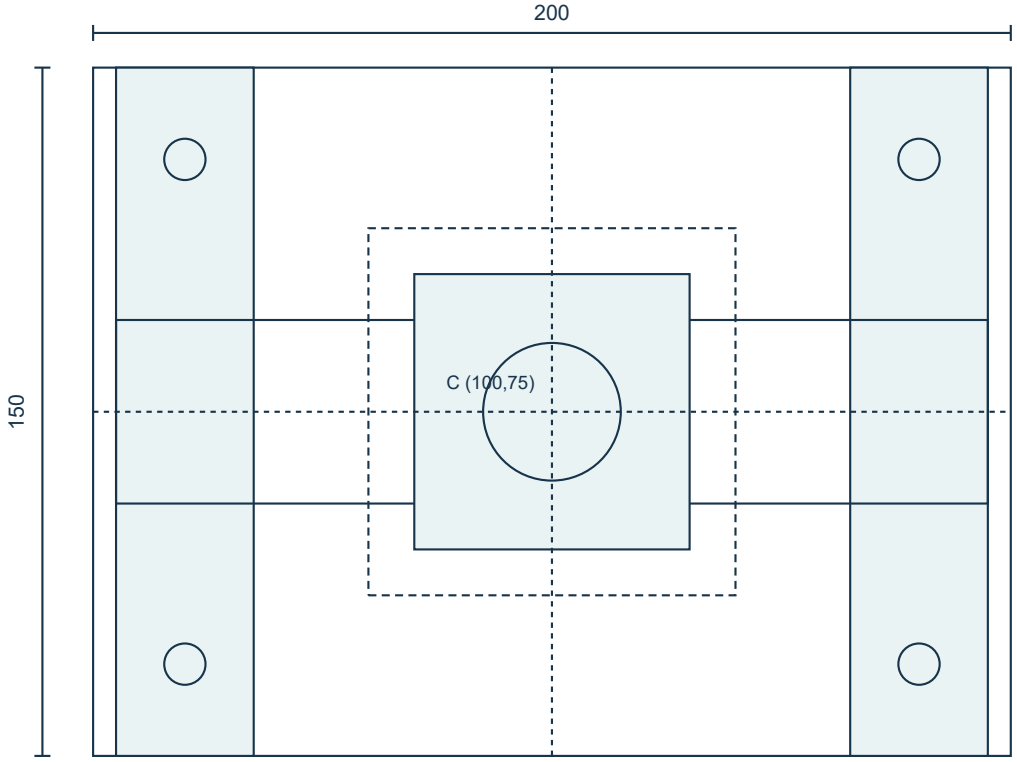
The acoustic opening must clear the port diameter plus twice the alignment uncertainty. A nominal 2 mm bore is only a starting proposal. A 1 mm annular gasket is proposed around the port, leaving the hole open. Glue and fasteners must not enter it. Compression stops and insulated edge supports will limit PCB bending. This gasket is distinct from the 3 mm phantom layer. [6]

Record tube/cavity geometry, gasket compression and orientation. The INMP441 typical acoustic lower corner is about 60 Hz; the 20-60 Hz risk remains open. Its internal digital high-pass corner is a different specification. A seal alone does not prove low-frequency suitability. [5]

4) A3 - Dimensioned phantom proposal

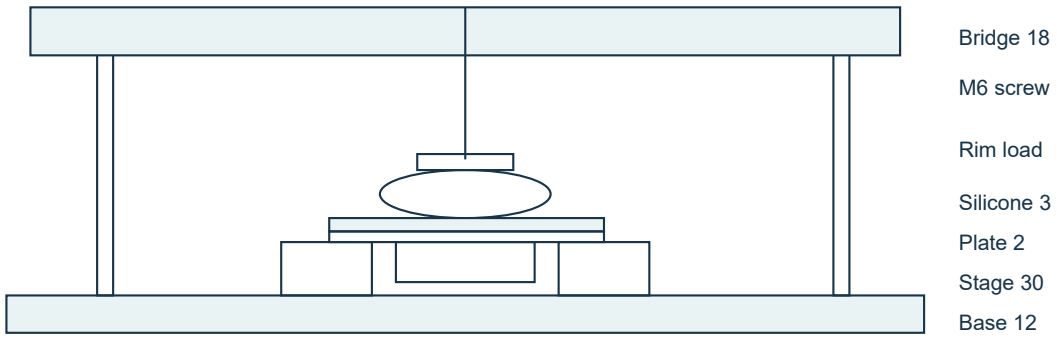
All dimensions are design proposals, not measurements of received parts. A removable prototype was defined. Fabrication must wait for the actual-part fit checks on page 5.

Top view (mm); origin = front-left of base



C: exciter / head axis. Ø9 post holes: (20,20), (180,20), (20,130), (180,130).

Front section - vertical stack (mm; schematic)



The base thickness was increased from the Week 4 proposal of 6 mm to a nominal 12 mm to provide more material for fixture attachments; stiffness remains untested. M8 posts set coarse height; the central M6 screw and swivel foot load the head rim. The microphone cassette is outside the loading path.

5) Build list and ten-repeat method

Proposed part	Nominal size / quantity	Condition
Plywood base	200 × 150 × 12 / 1	Flat base; four Ø9 holes as drawn.
Rails / bridge	30 × 150 × 18 / 2; 190 × 40 × 18 / 1	Rail centres x=20/180; bridge y=75. M6 connections at (20,75)/(180,75).
Posts / loading	M8 × 180 / 4; M6 × 100 / 1	Nuts/washers, captive M6 nut, swivel foot, rim collar; verify travel.
Stage / drive plate	80 × 80 × 30 ring; opening 40 × 40; plate 60 × 60 × 2	Exciter centred underneath; consistent edge support and removable clips.
Coupling layer	Silicone 60 × 60 × 3	Same-grade 2 and 5 mm alternatives; gel only in a sealed thickness-controlled pouch.
Other mounting	Cassette, tube, gasket, clamps	M3 screws, insulated supports, cable ties and position stops.

Fit conditions (mm): exciter body-plus-lead height ≤25 within the 30 mm stage retains 5 mm clearance; otherwise raise the stage/bridge. Confirm the head fits the 60 mm coupon without clip interference. Rail underside height is provisionally adjustable over 70-120 mm; measured head/foot sizes determine the setting. The 2 mm acrylic drive plate requires crack, deflection and resonance checks. Receipt of these mechanical items is unconfirmed.

Ten independent remounts - all results PENDING

Use the same coupon, orientation, tube route, waveform, level and acquisition profile. Each run requires removal, replacement against stops, force adjustment and the same recorded settling time. Initial 1 N force is an engineering proposal, not an accepted limit. Use a force gauge or tared scale supporting the whole stage; screw turns do not establish force. Agree force tolerance and settling time before tests.

Runs 01-10 retain before/after force, thickness, mount ID, raw-file ID and anomalies. Fixed-mount repeats must be reported separately. Record silicone grade/hardness and measured thickness; no material has been acoustically qualified.

Availability / potential missing items: verify chestpiece, tube, gasket, silicone, base/rails, fasteners, force gauge or scale, caliper, multimeter, regulated supply, oscilloscope, function generator and PC audio output in lab stock before procurement. Their absence has not yet been confirmed; only unresolved availability is recorded.

6) First low-level excitation procedure

PC analogue output → PAM8403 → 30 mm / 12 Ω exciter → drive plate + silicone → chestpiece → INMP441. The amplifier drives the exciter; it is not the microphone AFE. If no PC analogue output is available, an available USB audio device/DAC must be verified.

BTL connection: use L+ and L- of one complete channel. Neither terminal is ground. Do not connect either to ground, another output channel or an oscilloscope ground clip. Source ground belongs only to the amplifier input reference. Output measurements require an appropriate differential/isolated method and verified instrument limits. [7]

Step	Planned procedure / stop conditions
1 - Unpowered	Inspect labels, polarity, soldering and continuity. Confirm exact exciter model, rated power and nominal impedance. DC resistance is not AC impedance.
2 - Supply	Verify the actual board supply/input circuit. Use regulated power within rating; current limit depends on the verified load/board. Nominal 5 V is proposed. Begin with PC and amplifier levels at minimum.
3 - Connect	Connect the exciter before power. Do not parallel outputs or drive the exciter directly from the PC jack. Secure the assembly and wiring.
4 - Start	Apply a short low-level 100 Hz sine. Increase only until a usable response is observed. Stop for heating, buzzing, loose contact, excessive motion, current limiting or clipping.
5 - Extend	Only after the initial check, proceed to the frequency list. Do not increase low-frequency drive simply to compensate for microphone attenuation. Check fixture and ratings each time.
6 - Record / change	Set volume to zero and remove power before rewiring. Retain waveform, settings, supply/current observations, wiring photo and aborted-run notes. All steps: PENDING.

No maximum safe drive voltage was assigned because the exact exciter power rating is unconfirmed. The 30 mm / 12 Ω label and amplifier headline power do not prove 20-500 Hz performance or a safe displacement limit. No initial drive test was performed.

7) A4 - ECM academic comparison path

The main INMP441 → I²S → ESP32-S3 path requires no external microphone AFE. Kingstate KECG2740PBJ remains the academic comparison candidate. Replacing PUI with INMP441 because of supplier lead time does not cancel the ECM or analogue-noise task. [1, 4, 5]

Required chain and MCP6022 role

ECM → verified bias/load → AC coupling and bias reference → gain/filter → ADC → digital record. Kingstate uses a 2 V / 2.2 kΩ datasheet test condition; this is not an approved complete circuit. MCP6022 is a dual op-amp for proposed amplification/filtering or reference buffering, not an ADC or exciter driver. Its 2.5-5.5 V supply range must be reconciled with the whole circuit. [8, 9]

Route	Research / approval required
ESP32-S3 internal ADC	Pin/unit, calibrated range and nonlinearity, sample timing/loss, effective noise/resolution, anti-alias filter and digital interference. Nominal bit count is not effective audio resolution. [10]
External audio ADC	Input full scale/common mode, analogue supply, clocks, format, low-frequency poles, module coupling capacitors, availability and firmware. PCM1808 is a research example, not an ordered choice. [11]
Common analogue design	Gain from min/max inputs and headroom; bias, coupling poles, filtering, tolerances, decoupling and stable op-amp loading. Component values and ADC selection remain open.

Planned experiments - NOT YET MEASURED

After schematic review: measure DC operating points, then AFE gain/bandwidth with an attenuated, bias-compatible known signal. Measure terminated-input noise at an explicitly approved analogue test point while preserving the intended bias. Do not short an unknown powered input or a digital microphone pin. Compare ECM and INMP441 using the same phantom, coupling, stimulus and recording settings.

The advisor must confirm deferred Week 5 timing, ECM scope, the schematic/test-point definition and acceptance criteria before this path is called complete.

8) A5 - Week 6 test checklist

All results are **PENDING**. ESP32 receipt removes that item's delivery blocker; it does not establish a working chain. Each test requires its own verified prerequisites.

ID / status	Action	Prerequisite / retained evidence
W6-A01 PENDING	Power / wiring	Exact board pinout, grounds, shorts, polarity and measured rails; photo. No 5 V on INMP441 power/signals.
W6-A02 PENDING	First I²S capture	Agree Module B profile; verify clocks/channel, sign extension, sample alignment, timebase and loss count. Raw data + firmware version.
W6-A03 PENDING	Quiet baseline	Fixed mounting; amplifier on/off separately. Duration, filtering, RMS/PSD and environment. System noise, not isolated microphone self-noise.
W6-A04 PENDING	50 / 100 / 200 / 500 Hz	Approved low-level drive; fixed source/mount. Fundamental, harmonics, peaks and anomalies with actual drive settings.
W6-A05 PENDING	20 / 30 / 40 / 50 / 60 Hz	Controlled drive; track source/amplifier/exciter behaviour. An independent reference is needed to isolate microphone attenuation.
W6-A06 PENDING	Contact / mounting	Change one variable: remount, force or same-grade thickness. Ten-remount method on page 5. Retain valid and flagged runs.
W6-A07 PENDING	S1/S2 reference PCG	Use the team-selected dataset/file; retain segment, sample rate, scaling and playback path. No new dataset was selected here.
W6-A08 PENDING	ECM gain / bandwidth	Reviewed AFE/ADC, correct generator attenuation/bias and verified instruments. Do not inject a generator signal into I²S.

Last documented Module B baseline [4]: nominal 48 kHz, 64 BCLK periods/frame (3.072 MHz), proposed GPIO4/5/6 and 3.3 V microphone supply. This historical handoff is not a new pin approval. Ozan must confirm actual board mapping, firmware and USB routing before connection.

9) Academic alignment and handoff

Original Week 5: build the first breadboard front end and measure shorted-input noise [2]. This report does not fulfil that physical experiment. The new brief defers unapproved analogue work; revised timing and scope require advisor confirmation. The Week 5 milestone is not declared passed. The original Week 6 gain/bandwidth task is retained as W6-A08.

Task	Provided preparation	Remaining evidence
A1	Receipt table and inspection method	Counts, markings, condition and photos.
A2	Dimensioned cassette and seal rules	Actual sizes; assembly/fit check.
A3	Dimensioned frame, materials, repeat/safety plans	Availability; build and response tests.
A4	AFE/ADC research and experiment plan	Advisor decision; schematic; analogue noise.
A5	W6-A01 to A08 prerequisites	Physical records and reproducible analysis.

Open decisions before integration

Confirm the ECM ADC/AFE and approved noise-test point. Agree bandwidth, noise, distortion and clipping thresholds; the proposed Week 4 6 dB headroom remains unapproved. Confirm the low-frequency reference method, actual exciter rating, available instruments, chestpiece, silicone, frame and playback output. Confirm Module B interface and Module C analysis inputs. This report closes no shared decision.

Reproducible records

Retain raw signed samples, full-scale definition, format, firmware commit, fixture revision and settings. State bandwidth/DC removal for noise RMS; report clipping counts and dropped samples. A response normalised to a declared 100 Hz reference remains a whole-chain result unless excitation is independently calibrated. Simulation must never be labelled measured.

GitHub delivery and individual contribution

Target: docs/sdp/week-05/submissions/module-a-kaan/. Update W05_A_breadboard-front-end-bench-test.md with design, measured and blocked sections; add mechanics, inventory and test preparation via an own-account PR. This PDF does not claim upload or merge. Kaan’s design revisions, inspections and analysis contributions should remain traceable under the 70% team / 30% individual assessment. [3]

10) Sources and limitations

Bracketed numbers identify these sources. Manufacturer specifications do not qualify an unidentified breakout board or the assembled phantom. Web sources were reviewed on 26 October 2026.

[1] Supplied Week 5 coordination brief and later Module A receipt/measurement-status confirmation, 26 October 2026.

[2] Advisor: SDP4_Digital_Stethoscope_Plan_TR-1.docx. Semester 1 Module A Weeks 4-6 and Week 5 milestone.

[3] SDP_Assessment_Criteria_for_Students.docx. Evidence, reproducibility and team/individual assessment.

[4] Week 4 report, rev. 1.1 (19 October 2026), reported order and PR #28 handoff. Historical record, not new receipt evidence.

Open source document

[5] InvenSense/TDK: INMP441 datasheet, DS-INMP441-00, rev. 1.1. Acoustic response and I²S.

Open source document

[6] InvenSense: AN-1124, Recommendations for Sealing InvenSense Bottom-Port MEMS Microphones.

Open source document

[7] Diodes Incorporated: PAM8403 datasheet. BTL output, supply and application guidance.

Open source document

[8] Kingstate: KECG2740PBJ manufacturer specification distributed by Farnell.

Open source document

[9] Microchip: MCP6021/2/3/4 datasheet, DS20001685E. Amplifier function and supply.

Open source document

[10] Espressif: ESP32-S3 ESP-IDF ADC documentation. Use the project's actual SDK version.

Open source document

[11] Texas Instruments: PCM1808 datasheet. External ADC research example only.

Open source document

Boundary: dimensions, layer alternatives and loading settings are engineering proposals. No fabrication fit, calibrated noise floor, frequency-response PASS, full requirements freeze, hardware milestone completion or diagnostic performance is claimed.